

Functional Requirement

- 1.URL Shorten
- 2.Redirect to Original URL
- 3.User Accounts
- 4.URL Validation
- 5.Delete URL

Non-Functional Requirement

- 1.High Availability
- 2.Scalability
- 3.Fast Response Time
- 4.Reliability
- 5.Consistency

Estimation

1. Total Number of URLs

URLs/day = 10,000

Days in 20 years = 20×365
= 7,300

Total URLs
= $10,000 \times 7,300$
= 73,000,000 URLs

Total URLs = 73 million

2. Database Storage

Each URL record $\approx$ 300 bytes

$73,000,000 \times 300$
= 21,900,000,000 bytes
 $\approx$ 21.9 GB

Including indexes and database overhead ($\sim 2\times$):

$21.9 \times 2 \approx 44$ GB (So approx 50GB)

3. Read Traffic

Let's say each shortened URL is accessed 10 times on average.

$10,000 \text{ URLs/day} \times 10 \text{ clicks}$
= 100,000 redirects/day

Requests per second:

$100,000 / 86,400$
 ≈ 1.16 requests/sec

4. Cache Requirement

Let's say we cache the most popular 1 million URLs.

Each cache entry $\approx$ 300 bytes

$1,000,000 \times 300$
= 300 MB

= 512 MB (Approx)

Architecture

